

FLAT ISLAND LIGHT, AK, USA

WMO: 994720

Lat: 59.332N

Lon: 151.995W

Elev: 18

StdP: 101.11

Time Zone: -9.00 (NAK)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-11.4	-9.3	-15.3	1.0	-8.3	-14.0	1.1	-7.3	19.9	-2.9	18.2	-2.8	11.1	280	0.790

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.4	16.6	13.0	15.3	12.3	14.5	11.9	13.7	15.7	13.0	14.6	12.5	14.1	3.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.6	9.1	14.4	12.0	8.8	13.7	11.6	8.5	13.3	38.3	15.9	36.7	14.8	35.3	14.1	15.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.6	14.7	12.7	DB	-13.2	18.8	2.0	1.2	-14.6	19.6	-15.7	20.3	-16.8	21.0	-18.3	21.9	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.4	0.4	0.1	0.3	3.0	7.1	9.7	11.9	12.5	10.4	6.4	1.9	0.7		
	DBStd	5.53	4.64	3.72	3.74	2.53	1.90	1.69	1.61	1.40	1.92	2.56	3.91	4.14		
	HDD10.0	1866	299	276	299	209	92	25	2	1	18	112	244	288		
	HDD18.3	4720	557	509	558	459	348	260	199	182	239	368	494	546		
	CDD10.0	186	0	0	0	0	2	15	62	77	29	2	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH27.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	5.4	7.0	6.8	6.3	4.6	3.6	3.4	3.4	3.6	5.0	6.2	7.9	7.7		
(15) Precipitation	PrecAvg	629	54	47	32	28	21	28	45	67	95	71	67	75		
	PrecMax	871	181	156	71	97	55	50	92	114	169	207	197	126		
	PrecMin	382	1	7	3	5	7	7	14	7	31	30	8	10		
	PrecStd	99	37	32	20	21	11	12	20	26	36	35	47	33		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.1	7.7	7.2	10.1	14.5	17.1	18.7	18.0	16.0	13.5	10.4	9.1		
		MCWB	6.6	5.4	5.1	6.9	9.1	11.6	13.5	14.3	12.9	10.4	7.7	6.9		
	2%	DB	8.0	6.0	6.3	7.8	11.9	14.4	16.4	16.6	14.4	11.4	8.9	7.6		
		MCWB	5.7	4.1	4.5	5.6	7.7	11.1	12.9	13.4	11.6	9.0	6.7	5.2		
	5%	DB	6.8	5.2	5.6	6.9	10.6	13.0	15.2	15.3	13.6	10.3	7.9	6.5		
		MCWB	5.0	3.5	4.0	5.0	7.6	10.5	12.3	12.6	11.2	8.2	6.0	4.7		
	10%	DB	5.7	4.5	4.8	6.0	9.7	11.9	14.2	14.6	12.8	9.6	6.8	5.6		
		MCWB	4.3	3.1	3.4	4.3	7.3	9.8	11.9	12.2	10.7	7.6	5.3	4.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.9	5.6	5.8	7.1	9.8	12.8	14.1	14.8	13.5	10.7	8.3	7.2		
		MCDB	9.0	6.8	6.7	9.4	12.8	15.0	17.1	16.9	15.5	12.8	9.5	8.8		
	2%	WB	6.0	4.7	5.1	6.1	8.9	11.7	13.4	14.0	12.2	9.6	7.2	5.8		
		MCDB	7.7	5.7	6.0	7.7	10.9	13.6	15.7	15.9	13.6	10.9	8.6	7.0		
	5%	WB	5.3	3.9	4.5	5.2	8.3	10.8	12.8	13.3	11.7	8.6	6.4	5.1		
		MCDB	6.6	4.9	5.5	6.5	10.0	12.9	14.7	14.9	13.0	9.6	7.8	6.2		
	10%	WB	4.5	3.3	3.8	4.6	7.7	10.3	12.3	12.8	11.1	7.9	5.5	4.3		
		MCDB	5.7	4.4	4.9	5.9	9.5	11.9	14.0	14.2	12.5	9.1	6.9	5.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.3	3.4	3.4	3.4	3.5	3.5	3.7	3.4	3.0	2.8	3.3	3.3		
		MCDBR	4.4	3.9	3.5	4.4	5.0	5.4	5.2	4.6	3.9	3.8	4.0	4.2		
		MCWBR	3.0	3.0	2.5	2.6	2.6	2.7	2.6	2.5	2.6	2.7	3.3	2.9		
	5% WB	MCDBR	4.3	3.7	3.1	3.8	4.4	4.8	4.6	4.0	3.6	3.3	3.8	3.8		
		MCWBR	3.1	3.0	2.6	2.8	2.7	2.7	2.6	2.4	2.7	2.6	3.3	2.9		
(40) Clear-Sky Solar Irradiance	taub		0.265	0.310	0.319	0.360	0.378	0.366	0.369	0.362	0.340	0.322	0.284	0.272		
	taud		2.259	2.332	2.395	2.319	2.325	2.448	2.488	2.501	2.519	2.400	2.194	2.041		
	Ebn at Noon		554	708	824	841	850	869	857	837	798	685	495	352		
	Edh at Noon		41	66	83	106	113	102	96	88	74	61	43	35		
(44) All-Sky Solar Radiation	RadAvg		0.33	0.87	2.18	3.54	4.85	5.35	4.74	3.78	2.48	1.21	0.47	0.20		
	RadStd		0.04	0.15	0.34	0.49	0.49	0.45	0.43	0.48	0.36	0.13	0.08	0.02		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	+0.84	+2.34	+2.18	N/A	N/A	N/A	N/A	-225	-287	N/A	N/A

Nomenclature: See separate page